DDOT | Tableau Assessment

Executive summary

The Associate Director of the Public Space Division requested a Tableau workbook measuring average business days to approve new construction permits issued during FY2026, including comparisons by permit type and an additional meaningful segment. We created a permit-level descriptive analysis using Python to extract, validate and transform the DDOT TOPS ArcGIS REST feed, and Tableau to display overall approval time, construction activity comparisons, and monthly application cohorts.

Metric	Preliminary finding
Extracted location records	28,277
Unique tracking numbers	5,378
Overall mean business days	11.40
Median / 90th percentile	1 / 40 business days
Main output	Three worksheets, optional permit count KPI, one dashboard

Interpretation caution: the measure uses LastApprovalDate provisionally, and monthly cohorts are subject to selection bias because only permits issued during FY2026 appear in the sample.

1. Why Python and Tableau were used together

Tool	Responsibility	Reason
Python / requests	ArcGIS REST data retrieval	Automated extraction, pagination, retry handling and request length control
Python / pandas	Entity-level QA and data transformations	Reproducible deduplication review, dates, groups and audit outputs
Python / NumPy	Business-day interval calculation	Explicit weekend and DC holiday exclusion
Tableau	Visual analysis and presentation	Interactive worksheet aggregation, labels, tooltips and director-facing dashboard

2. Full processing history: from start to Tableau

Extract FY2026 candidate records

The script step1_extract_audit.py queried server-side matching record counts and OBJECTIDs, then retrieved selected attributes in batches. Initial requests listing about 2,000 IDs produced HTTP 414 (Request-URI Too Long). The corrected script uses 150 OBJECTIDs per GET request, with retries, to avoid oversized URLs. It saves the candidate CSV and audit_summary.json. Extracted count: 28,277 rows.

Identify duplicated location records

The 28,277 rows contained 5,378 distinct tracking numbers. A review found 22,808 exact repeated records when excluding OBJECTID and 36 permits associated with multiple wards. Raw-row weighting gave an existing DaysToApprove average of 26.35 days, while permit-level averaging gave 16.39, demonstrating why repeated rows distort metrics. The workflow created fy26_permit_level_preliminary.csv and permit_ward_relationships.csv. No separate executable review/deduplication script was retained, so full reproducibility from the raw file is a documented gap.

Calculate business-day approval time

The script step2_business_days.py calculates ApprovalBusinessDays from ApplicationDate through LastApprovalDate using a Monday–Friday NumPy calendar excluding explicitly coded observed DC holidays. It excludes the application date, includes a valid approval weekday, and defines same-day approvals as zero. It generates a per-permit CSV and audit. For the 5,378 records, the computed mean was 11.40 business days, median 1, 90th percentile 40, and 107 records exceeded 90 business days. No invalid application-to-approval dates were identified.

Cross-check source duration fields

Compared DDOT DaysToApprove to (a) calendar-day difference and (b) calculated business days. Calendar-day differences matched 4,939 source values; 439 did not. Business-day values matched 3,320 and differed for 2,058. A 37-case exception file was prepared. These comparisons indicate the source field cannot be assumed equivalent to the required custom metric, but do not prove source records are erroneous. Independent TOPS permit-history confirmation remains outstanding.

Derive construction activity groups

step4_classify_permits.py processes TypeDetailNames as comma-separated activities with 'Category; Description' patterns. It extracts distinct category prefixes; single-category permits receive the family, while permits with multiple families receive 'Multiple activity categories'. This provides mutually exclusive grouping for permit-level comparison. The output includes a final preliminary CSV, category and monthly summary files and a category audit.

Connect the main CSV in Tableau

Tableau Desktop connects to tableau_permit_analysis_preliminary.csv using Text File. TrackingNumber should be text; ApprovalBusinessDays numeric; ActivityCategory text; ApplicationMonthStart date. Summary CSVs were retained for independent validation rather than brought into the Tableau data model.

Category worksheet

Category worksheet: ActivityCategory on Rows and AVG(ApprovalBusinessDays) on Columns, Bar marks, descending average, numeric labels and distinct tracking count in tooltip. Mega Projects and PS Rental have one permit each and were omitted from the category chart only. Monthly worksheet: continuous MONTH(ApplicationMonthStart) on Columns, AVG(ApprovalBusinessDays) on Rows, Line marks and distinct count in tooltips. An overall text KPI uses AVG(ApprovalBusinessDays), and a second KPI uses COUNTD(TrackingNumber).

Build and refine the dashboard

A tiled dashboard includes a title, two KPI worksheets side by side in a Horizontal Container, the category bar chart, monthly trend, key findings and a methodology note. Iterative formatting fixed oversized containers, truncated labels, incorrectly aligned KPI text, scrollbars, and text objects compressing charts. The dashboard screenshot development was ongoing; final spacing, chart visibility and filter behavior must be checked in the saved Tableau file.

3. Category findings

Activity family	Permits	Average business days
Excavation	4,538	10.89
Multiple activity categories	399	13.70
Over Head Work	202	8.39
Paving	97	18.77
Fixture	94	22.55
Projections	26	12.69
Landscaping	20	23.90
Mega Projects	1	0.00
PS Rental	1	0.00

Excavation is the dominant group: 4,538 of 5,378 permits (~84.4%). Fixture and Paving have higher observed category averages (22.55 and 18.77) than Excavation (10.89). These comparisons do not establish causes of differences. Small categories should not be overinterpreted.

4. Monthly cohort results and interpretation

Application month	Permits	Average business days
2025-10	178	26.38
2025-11	82	11.96
2025-12	33	82.48
2026-01	37	76.30
2026-02	90	61.43
2026-03	179	48.49
2026-04	110	47.98
2026-05	324	35.44
2026-06	657	14.59
2026-07	1,586	3.91
2026-08	1,102	2.20
2026-09	1,000	0.91

A pronounced observed decline across FY2026 is not evidence by itself that operations became faster. Only permits issued by 2026-09-30 are included; later-submitted cohorts have less opportunity for long-duration cases to enter that issued-permit population. This creates a serious cohort comparability limitation.

5. Quality checks, decisions, issues and constraints

- FY26 date bounds are explicit and inclusive/exclusive to avoid ambiguity at midnight.
- Current issued/expired status is retained for review rather than used to discard historically issued permits.
- Unit of analysis is one tracking number, while additional ward links are retained separately.
- The custom measure is independent of DDOT DaysToApprove and accounts for stated weekend/holiday exclusions.
- Activity grouping is mutually exclusive; multi-family permits form a dedicated category.
- Category groups with only one observation are excluded from the chart but included in overall KPIs.
- The source field definition of LastApprovalDate, same-day convention, agency closure calendar and new/reissue eligibility still need authoritative validation.
- CSV snapshot is from 2026-10-14 and the source may refresh daily.
- The final packaged workbook was not available for programmatic QA while this report was prepared.

6. Reproduction file dependencies

Python 3 packages: requests, pandas and numpy. Execute step1_extract_audit.py to extract candidate records; use the provided fy26_permit_level_preliminary.csv to run step2_business_days.py; then run step4_classify_permits.py with the Step 2 output accessible at its expected path. The raw-row-to-permit deduplication processing was not provided as a standalone executable script and should not be represented as fully automated end-to-end reproduction.